

Design and Implementation of an ATmega32-Based Mathematical Equation Plotting Robot

1. Introduction

Embedded systems are integral to modern automation, robotics, and precision control applications such as CNC machines, plotters, and industrial robots. These systems require tight integration between hardware and software to achieve deterministic and reliable performance. This project proposes the design and implementation of a low-cost, bare-metal embedded system using the ATmega32 microcontroller that is capable of plotting mathematical equations on a two-dimensional surface. The system will receive equation symbols via serial communication, process them using embedded firmware, and convert them into coordinated mechanical motion using stepper motors and a pen actuation mechanism. The project emphasizes register-level programming, real-time control, and hardware–software co-design, providing strong practical exposure to industrial embedded system development.

2. Motivation

Many academic and hobbyist robotics projects rely on high-level platforms such as Arduino, which abstract critical low-level details of microcontroller operation. While these platforms simplify development, they limit understanding of fundamental concepts such as memory mapping, timers, interrupts, and peripheral control. The motivation of this project is to bridge this gap by developing a complete system using the ATmega32 microcontroller in a bare-metal environment.

By focusing on plotting mathematical equations instead of free-form handwriting, the project maintains manageable complexity while still demonstrating advanced concepts such as motion planning, deterministic control, and modular firmware design. This approach ensures both technical depth and feasibility within an academic timeframe.

3. Objectives

The main objectives of this project are:

- To design and implement a bare-metal embedded system using the ATmega32 microcontroller
- To develop register-level drivers for UART, GPIO, timers, and PWM
- To receive mathematical equation symbols through serial communication
- To convert equation symbols into predefined motion trajectories
- To control a two-axis stepper motor mechanism for accurate plotting
- To design a pen actuation system using a servo or solenoid mechanism
- To integrate power management, firmware, and mechanical subsystems into a reliable working prototype

4. Proposed Methodology

The system will be developed using a modular hardware and software architecture. Equation symbols will be transmitted from a computer to the microcontroller through an FTDI-based UART interface. The ATmega32 will parse the received symbols and map them to predefined stroke patterns stored in program memory. A motion planning algorithm will convert these strokes into step and direction signals for the motor drivers controlling the X and Y axes. Pen lifting and lowering will be handled using a servo motor or solenoid controlled via PWM. The firmware will be written in embedded C using AVR-GCC and will follow a layered structure separating low-level hardware drivers from application-level logic. Timer interrupts will be used to ensure accurate motor stepping and real-time behavior. The entire system will be powered using a lithium-ion battery with appropriate voltage regulation and protection. The ATmega32 will process user inputs and generate appropriate control signals for the actuation system. Motor driver circuits will be interfaced with the microcontroller to control servo motors, DC motors, and a solenoid mechanism. Mathematical relationships governing motor speed, torque, step resolution, and timing will be derived and implemented to achieve precise and repeatable motion control, particularly for drawing mathematical equations.

Serial communication will be established using an FTDI USB-to-TTL module for debugging, monitoring, and system analysis during development. A dedicated programmer will be used to upload firmware directly into the ATmega32. The mechanical structure of the system will be constructed using gears, shafts, and 3D-printed supporting components designed to ensure proper alignment, stability, and smooth movement of the writing mechanism. The software will be developed in a modular manner using timer interrupts, PWM techniques, and efficient GPIO handling. System testing will be conducted incrementally, starting from individual module verification and progressing toward full system integration and performance evaluation.

Block Diagram

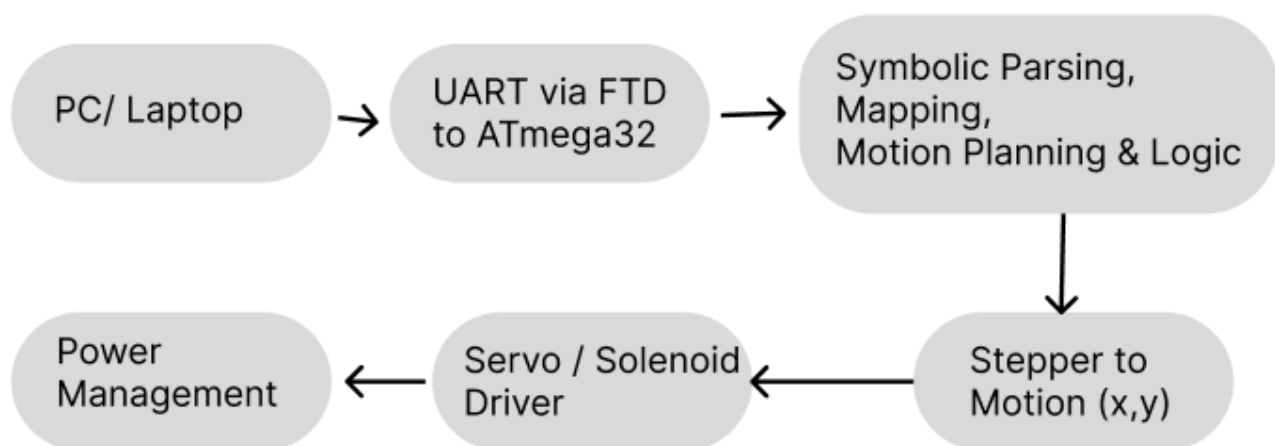


Fig 1: System Diagram

Circuit Diagram:

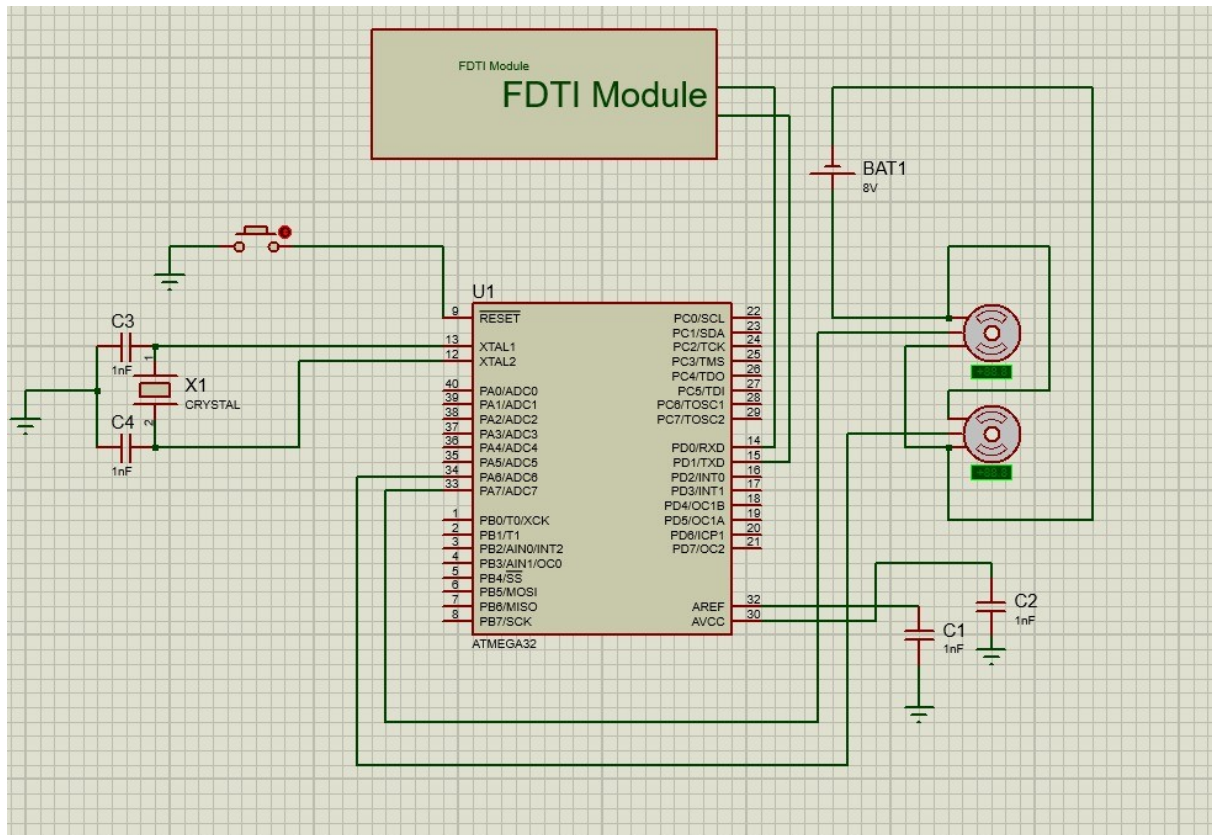


Fig 2: Expected Circuit Diagram in Proteus of Plotting Robot

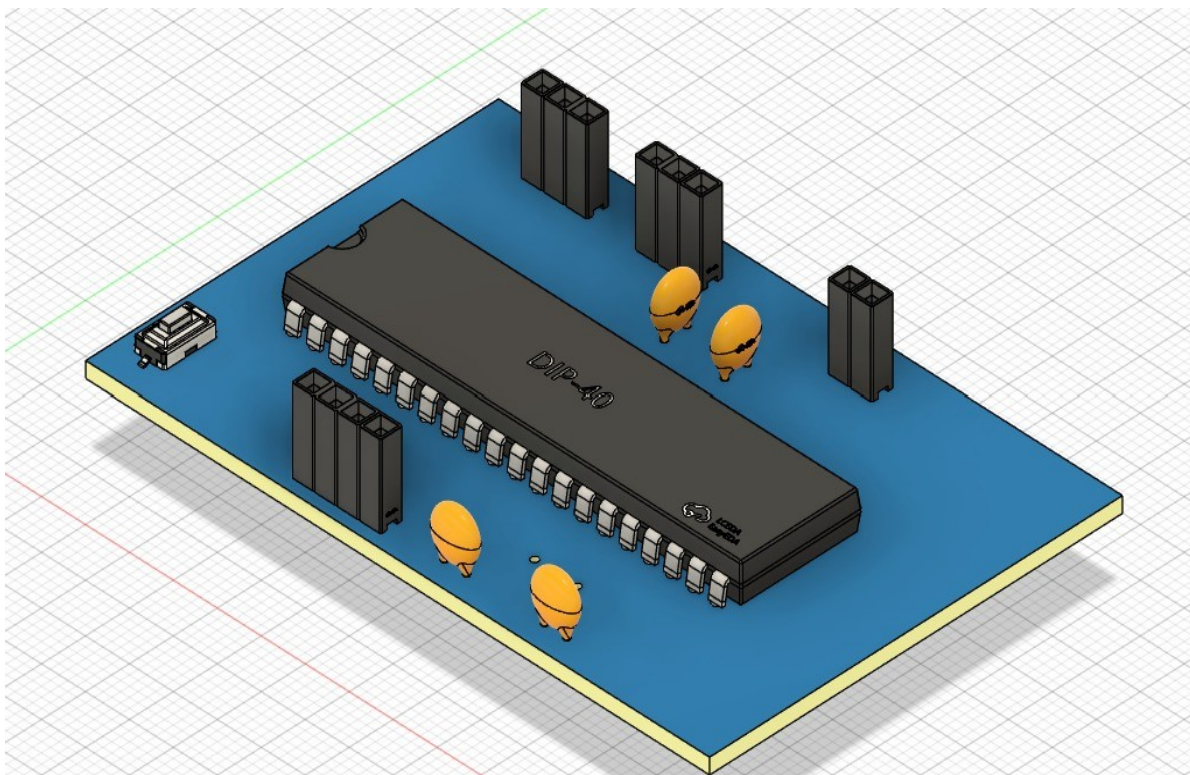


Fig 3: Design of the System in IC or Developed Board of Atmega32

5. Expected Outcomes

Upon successful completion of the project, the following outcomes are expected:

- A fully functional ATmega32-based robotic system capable of plotting basic mathematical equations
- Reliable UART communication between a PC and the embedded system
- Accurate two-axis motion control using stepper motors
- Stable pen actuation mechanism for drawing operations
- Modular and well-documented embedded firmware written at register level
- Practical understanding of real-time embedded system design and debugging
- A strong academic and professional portfolio project demonstrating embedded systems, automation, and robotics expertise

6. Societal Impact

The proposed project holds strong educational, technological, and socio-economic significance, particularly in the context of developing nations. By focusing on bare-metal microcontroller programming rather than ready-made platforms, the project enhances practical learning and develops critical engineering skills such as hardware debugging, low-level programming, and system-level problem solving. This approach helps bridge the gap between academic theory and real-world industrial requirements.

From a technological perspective, the project demonstrates how low-cost and widely available microcontrollers like the ATmega32 can be used to develop precise and reliable automation systems. The methodology and design principles can be extended to various applications including educational robotics, low-cost automation tools, and assistive writing or drawing systems. This promotes innovation using locally available resources and reduces dependence on expensive imported technologies.

The economic and social impact of the project lies in its emphasis on affordability and accessibility. The system provides a foundation for developing cost-effective automated solutions that can reduce repetitive manual work and improve productivity in small-scale applications. In the long term, the skills and knowledge gained through this project can contribute to the development of a technically skilled workforce capable of supporting advancements in robotics, automation, and intelligent control systems within the country.

7. Timeline

The project is planned to be completed within a structured academic timeline as outlined below:

Phase	Activities	Duration
Phase 1	Literature review, system planning, component selection	1 st , 2 nd week
Phase 2	ATmega32 setup, GPIO and UART driver development	3 rd week
Phase 3	Timer, interrupt, and PWM implementation	3 rd week
Phase 4	Stepper motor control and motion planning	3 rd week
Phase 5	Pen actuation system integration	4 - 5 th week
Phase 6	Equation symbol mapping and plotting logic	7 - 8 th week
Phase 7	System integration, testing, and debugging	9 th week
Phase 8	Documentation and final demonstration	10 th week

Table 2: Total Estimated Duration: 10 weeks

8. Budget

Item	Quantity	Estimated Cost (BDT)
ATmega32 Microcontroller	1	350
Crystal Oscillator & Passive Components (Resistors, Capacitors)		300
Voltage Regulator & Power Conditioning Circuit	1	300
FTDI USB-to-TTL Module	1	450
Servo Motors	2	1,000
Servo Motor / Solenoid Mechanism	1	400
Lithium-Ion Battery & Protection Circuit	1	300
Mechanical Structure, Gears & 3D Printed Parts		1,500
Wiring, Connectors & Miscellaneous Hardware		400

Table 3: Total Estimated Cost: (5000) Moderate and student-friendly.

9. References

1. M. A. Mazidi, *AVR Microcontroller and Embedded Systems*, Pearson Education
2. Atmel Corporation, *ATmega32 Datasheet*
3. Michael Barr, *Programming Embedded Systems in C and C++*
4. ExploreEmbedded.com – AVR Embedded Systems Tutorials
5. AVR Freaks Community Forum